IT Employee Attrition Analysis

(Simulated Data) - Project Documentation

Author	Santosh Chaurasia
Project type	Practice project
Tools	Python, SQL (SQL Server), Excel, Power BI
Dataset	Simulated data, 1,000 employee records

Note: The data in this project is simulated. I made it with Python using rules that I wrote myself. It is not from a real company, so the results do not show real employee behaviour. I made this project only to practice the full analysis steps.

1. About the Project

In this project I wanted to practice a full data analysis from start to end. I made a dataset of 1,000 IT employees and looked at which groups of employees leave the company more (attrition): by salary, overtime, experience, department, work mode, promotion and distance from home.

I used four tools: Python to make and check the data, SQL Server to find attrition rates, Excel for pivot tables and a dashboard, and Power BI for a second dashboard.

2. Goals

- Make a dataset about employees and whether they left.
- Use SQL to find which groups leave the most.
- Check the same things in Python with simple charts.
- Make a dashboard in Excel with pivot tables, charts and slicers.
- Make a dashboard in Power BI with KPI cards, charts and filters.

3. Tools Used

Tool	Used for
Python (Pandas, NumPy, Matplotlib)	Making the data, checking it and making attrition charts
SQL Server	Creating the table and 5 queries with CASE and GROUP BY
Excel	5 pivot tables, 5 pivot charts and 4 slicers
Power BI	4 KPI cards, 2 charts and 2 filters

4. Dataset

The file `it_employee_attrition_dataset.csv` has 1,000 rows and 9 columns. I checked it in Python and there are no missing values and no duplicate rows.

Column	Meaning
Employee_ID	Unique ID (1001 to 2000)
Department	IT, Sales, HR or Operations
Total_Experience_Years	Experience in years (1 to 12)
Salary_LPA	Salary in lakhs per year (3.9 to 21.29)
Years_Since_Last_Promotion	0 to 5 years
Work_Mode	WFH, Hybrid or WFO
Distance_From_Home_KM	Distance from home (2 to 44 km)
Overtime	Yes or No
Attrition	Yes if the employee left, No if the employee stayed

How I made the data: I used NumPy with seed 42 in the notebook `IT_Employee_Attrition_Data_Generation.ipynb`. These are the rules I used:

- Department (IT 45%, Sales 25%, Operations 20%, HR 10%), work mode (Hybrid 40%, WFH 30%, WFO 30%) and overtime (Yes 35%) are picked randomly. Experience, promotion years and distance are random numbers.
- Salary = experience x 1.4 + a random number between 2.5 and 4.5.
- Attrition starts with a 5% chance of leaving. Then I added extra chance: salary below 6 LPA (+15%), 3 or more years without promotion (+10%), overtime (+20%), WFO with distance above 25 km (+12%), experience 2 years or less (+15%), Sales department (+5%) and WFO (+5%).
- A random number decides if the employee leaves.

Because I set these rules myself, the analysis only confirms them, it does not discover them.

5. Steps I Followed

- Made the 1,000 employee rows in Python and saved them as a CSV file.
- Checked the data in Python (shape, info, missing values, duplicates).
- Created the database and table in SQL Server, loaded the CSV and ran 5 queries.
- Made 7 attrition charts in Python.
- Made pivot tables, charts and slicers in Excel.
- Made a one-page dashboard in Power BI.

6. SQL

I created the database `IT_Employee_Attrition_DB` and the table `it_employee_attrition`. In each query I used CASE to make groups and calculated the attrition rate (%), because a bigger group always has more people leaving, so the count alone can be misleading. I checked the results with Pandas and they match.

#	Query	Result
1	Attrition by Salary Bracket	Below 6 LPA: 117 employees, 60 left (51.28%). 6-12 LPA: 338 employees, 76 left (22.49%). Above 12 LPA: 545 employees, 121 left (22.20%).
2	Overtime vs Attrition	Overtime: 358 employees, 133 left (37.15%). No overtime: 642 employees, 124 left (19.31%).
3	Experience vs Attrition	0-2 years: 177 employees, 80 left (45.20%). 3-7 years: 399 employees, 84 left (21.05%). 8+ years: 424 employees, 93 left (21.93%).
4	Department-wise Attrition	Sales 32.67% (251 employees), Operations 27.64% (199), IT 22.56% (461), HR 17.98% (89).
5	Work Mode Attrition	WFO 31.45% (318 employees), WFH 26.37% (273), Hybrid 20.78% (409).

7. Python

IT_Employee_Attrition_Data_Generation.ipynb: makes the 1,000 rows with the rules in section 4, then checks the data and does a few quick comparisons. Employees who left have a lower average salary (11.53 LPA) than employees who stayed (13.01 LPA).

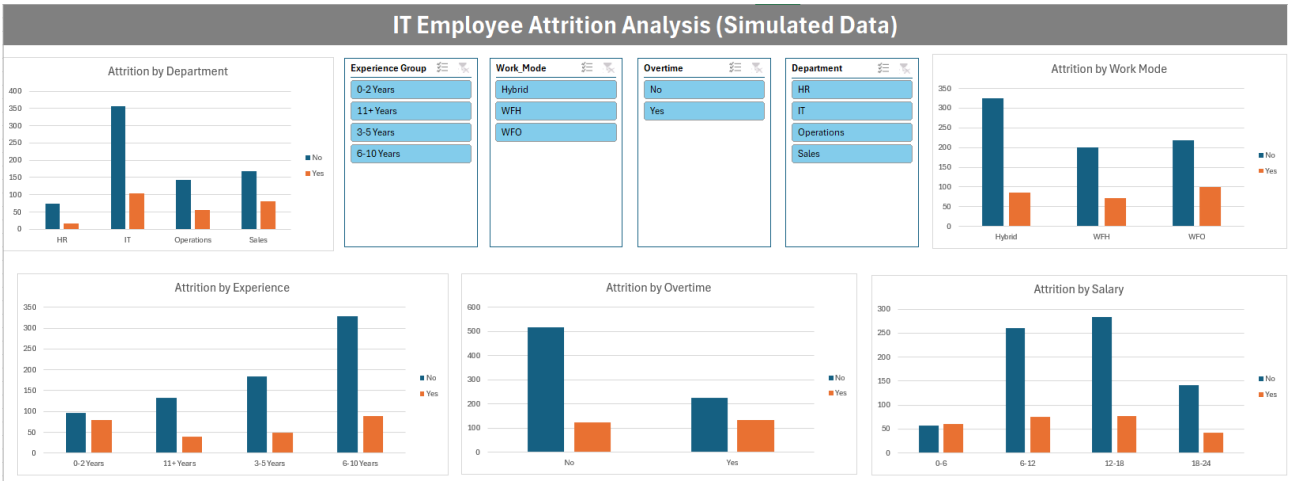
IT_Employee_Attrition_Analysis.ipynb: loads the CSV, checks missing values and duplicates, and makes a new column (1 if the employee left, 0 if not) so that the average gives the attrition rate. It then makes groups for experience, salary and distance, and makes bar charts of the attrition rate (%) by department, work mode, overtime, experience, salary, years since last promotion and distance from home. The charts are saved in a charts folder. At the end it checks overtime and experience together.

The overall attrition rate is 25.7% (257 out of 1,000 employees). The average experience is 6.54 years.

8. Excel Dashboard

The workbook `IT_Employee_Attrition_Analysis.xlsx` has 3 sheets: the data, a calculations sheet with 5 pivot tables, and the dashboard. The dashboard has 5 pivot charts (attrition by department, work mode, experience, overtime and salary) and 4 slicers (Experience Group, Work_Mode, Overtime and Department). I added an Experience Group column with a formula (0-2, 3-5, 6-10 and 11+ years). The title says it is simulated data.

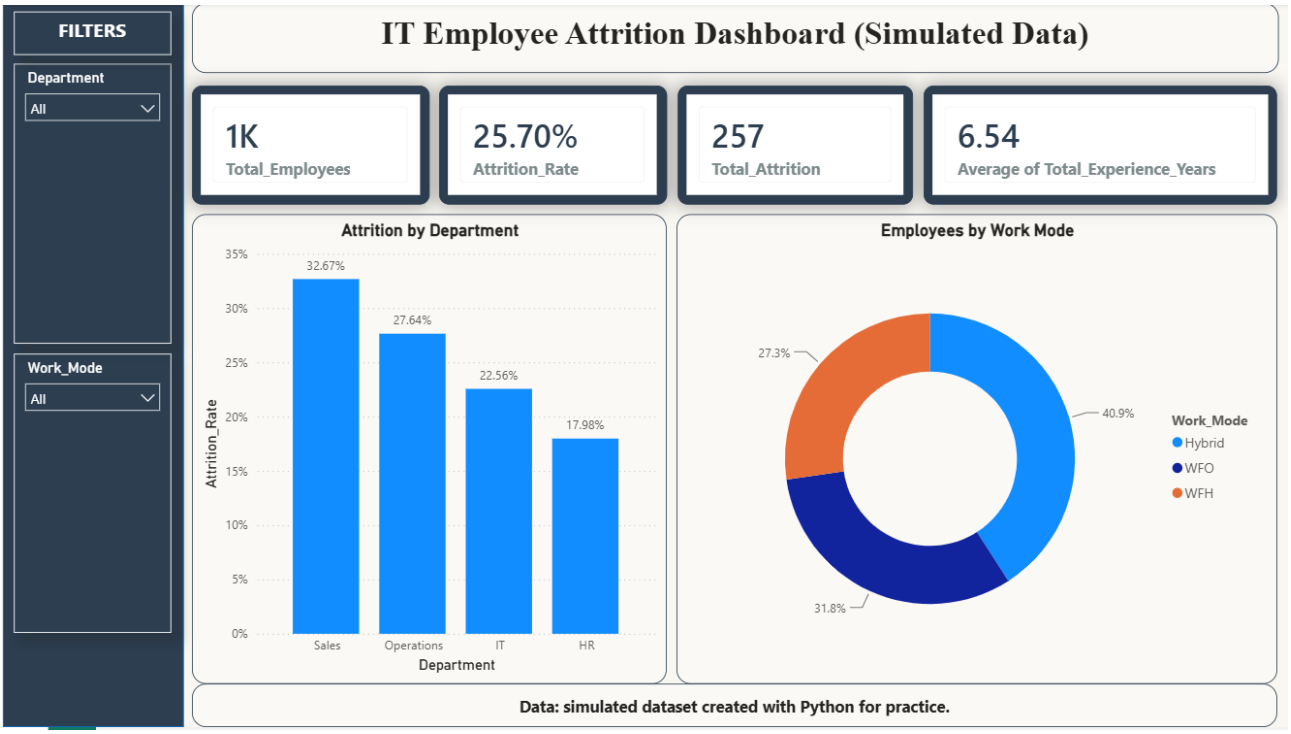
The Excel charts show the number of employees who stayed (No) and left (Yes). The Excel groups for experience and salary are a little different from the SQL groups, so the numbers are not the same as the SQL rates.



Excel dashboard (simulated data).

9. Power BI Dashboard

The Power BI dashboard has 4 KPI cards (Total Employees, Attrition Rate, Total Attrition and Average Experience), a column chart of attrition rate by department, a donut chart of employees by work mode, and filters for Department and Work Mode. The title and the footer say it is simulated data. The first card shows 1K, which is rounded; the real value is 1,000 employees.



Power BI dashboard (simulated data).

10. What I Found

These results are from the simulated data only.

- 257 out of 1,000 employees left, so the attrition rate is 25.7%.
- **Salary:** employees earning below 6 LPA leave the most (51.3%). The medium (22.5%) and high (22.2%) salary groups are almost the same.
- **Experience:** employees with 0-2 years leave the most (45.2%). Mid-level (21.1%) and senior (21.9%) are much lower. Salary depends on experience in this data, so these two results are

connected.

- **Overtime:** employees who do overtime leave more (37.2%) than others (19.3%). Entry-level employees with overtime have the highest rate (56.9%).
- **Department:** Sales has the highest rate (32.7%), then Operations (27.6%), IT (22.6%) and HR (18.0%). IT has the most employees leaving (104), but only because it is the biggest department.
- **Work mode:** WFO has the highest rate (31.5%), then WFH (26.4%). Hybrid is the lowest (20.8%).
- **Promotion:** attrition goes up after 4 and 5 years without a promotion (28.5% and 33.9%).
- **Distance:** distance has only a small effect (28.5% for 31+ km, about 23% to 25% for the others).

11. Limitations

- The data is simulated, so all these patterns come from the rules I wrote. The analysis only confirms them.
- HR has only 89 employees, so its result is less reliable.
- The Excel dashboard uses different groups than SQL and Python, and its charts show counts, not rates.
- I did not use any statistical tests, only simple counts and percentages.

12. What I Learned

- How to make a dataset with random choices and rules in Python.
- How to use CASE and GROUP BY in SQL Server to make groups and find rates.
- Why attrition rate (%) is better than the raw count when groups have different sizes.
- How to make pivot tables, slicers and a dashboard in Excel and Power BI.
- That results from simulated data mostly show the rules I put in it.

13. How to Run

- **Python:** open the notebooks in Jupyter or Colab (pip install pandas numpy matplotlib). The generator saves the file as `it_employee_attrition_data.csv`, but the analysis notebook reads `it_employee_attrition_dataset.csv` (the file in the DATA folder), so rename the file or change the name in the notebook.
- **SQL:** run `Create Database and Table.sql`, import the CSV into the table `it_employee_attrition` and run the query files.
- **Excel and Power BI:** open the files in the EXCEL and POWER BI folders.

Author

Santosh Chaurasia
M.Sc. Data Science student